

Priority	Design Requirements	Testing Procedure	Explanation	Why?
1.(High)	Safe	Kids will be observed using the gear wall with baby proofing material on the corners and minimal room for self-harm between cracks through observation of the aids or ourselves.	We need to ensure that the kids are safe and are unable to hurt themselves with the gear wall so that they can have a fun, stimulating interaction	Safety is the foundation of any product designed for children, especially those with special needs who may be more vulnerable to injuries. Ensuring zero risk of harm creates a trustworthy and worry-free environment for both children and caregivers.
2.(High)	Engaging	The kids will be observed using the gear wall to see if the gear wall is able to engage the kids for 5+ minutes through visual and physical stimulation through observation of the aids or ourselves.	We need to ensure that the kids are able to be stimulated for a set period of time so the aids may be given time for preparation in the next activity or helping other kids without worrying.	Engagement ensures children are motivated to interact with the gear wall. For special needs children, sustained engagement can support longer periods of focused learning and play, which are critical for development.
3.(High)	Conducive to Cause & effect Learning	The kids will be observed to see whether or not they understand that cranking the handle results in the gears turning and the lights turning on through observation of the aids or ourselves.	We need to ensure that the kids understand that their actions have results, in this case, the cranking results in the gears turning and the lights turning on.	Understanding cause and effect is a fundamental cognitive skill. It helps children grasp how their actions influence their environment, promoting problem-solving, logical thinking, and curiosity.
4.(High)	Accessibility	Each kid will be observed attempting to use the gear wall to see if they are able to use it or not through	We need to make sure that the gear wall is inclusive to all the kids with cognitive or physical disabilities.	Special needs children have diverse abilities, and an accessible design ensures all children can benefit from

		observation of the aids or ourselves.		the gear wall, regardless of physical, cognitive, or sensory challenges.
5.(High)	Fine motor skills	The kids will be observed how they grip the crank to see if they are able to use the crank, over the span of two weeks, kids will be assessed if their fine motor skills have improved from their base line through observation of the aids or ourselves.	The kids will need to be able to use their fine motor skills to be able to handle and crank, therefore having basic motor skills is key to using this gear wall.	Fine motor skills are crucial for everyday tasks like writing, buttoning clothes, and using utensils. The gear wall offers a playful way to practice and improve these skills.

Mr. Scheidly (Drexel University)

The metrics here are based on observation, which is perfectly legitimate, but you should consider ways to make them quantifiable so that you can set some concrete pass-fail criteria.

I'm not clear what is meant by statements to the effect of 'The children engage the gear wall', but perhaps this clarification exists elsewhere.

The safety section lacks specificity. Are there other dangers, such as running afoul of the crank or pinching fingers in the gears?

Safety is listed as a priority, but material wear and tear is not mentioned over time. What happens if the gears become loose, jam, or break?

Dr Jaimie Dougherty (Drexel University)

- cause/effect - but how will you know? Will you ask them to make the light turn on and see what they do? I think you need a clear protocol that you are following that clearly identifies if they understand.
- Fine Motor - is the handle made in such a way that it requires good fine motor? Just thinking about how often you can make something move with a bad grip, and get the job done, so trying to design a handle specifically to develop a specific grip seems like a really big ask. I think I'd focus more on if they need to adjust their grip often or something along those lines. But even with that, what

does it mean if this test fails? I think it's really important to always ask yourself how you would interpret success and failure into a future design revision.

Mr. Scheidly, a physics professor at Drexel University, told us that we should make our testing procedures more quantifiable as well as adding the possible dangers of pinching fingers into our safety section. He also adds that we should mention material wear and tear and asks what would happen if our gears were to become loose, jam, or break which informed us that we should put more focus into the safety of possible pinching of the students fingers in our design. Dr Jaimie, an engineering professor at Drexel University, mentioned that we should focus more on how often the students adjust their grip rather than developing a specific grip for individual students. She also questions what it means if our test fails. Lastly, she reminds us that it is paramount to always ask ourselves how we interpret the results of success and failure into future design revisions. Her feedback helped us better understand that the students don't really need specific handles to them; instead, we should be asking how the students grip changes. She also helped us question and really think about how we interpret success and failure into our design revisions.